

Seemarani G

Python Full Stack Developer | Voice & Teams Bot Engineer | Azure STT/TTS | Conversational AI
seemaranig2599@gmail.com | +91 93533 68540 | Bangalore, India | [linkedin.com/in/seemarani-g](https://www.linkedin.com/in/seemarani-g)

PROFESSIONAL SUMMARY

Python Full Stack Developer with 4+ years of hands-on experience building voice bots, real-time STT/TTS pipelines, and conversational AI systems for enterprise clients including Pfizer. Extensive experience with Azure Speech-to-Text, Azure Text-to-Speech, real-time transcription, and voice integrations. Delivered production AI systems using Python, FastAPI, Azure Cognitive Services, and Microsoft Teams-compatible audio/streaming architectures. Skilled in REST APIs, WebSocket streaming, React (basic), and Azure DevOps CI/CD in Agile/Scrum environments.

CORE SKILLS

Languages: Python · C# (Unity)

Voice AI & STT/TTS: Azure Speech-to-Text (STT) · Azure Text-to-Speech (TTS) · Azure Cognitive Services · Real-Time Transcription · Voice Integrations · ElevenLabs TTS · Streaming Inference · Chunked Audio Delivery

Microsoft Ecosystem: Azure Speech Services · Azure OpenAI (GPT-4) · Azure DevOps · Azure Cognitive Services · Microsoft Teams Audio/Voice Integration · Bot Framework (familiar) · Microsoft Graph APIs (familiar)

Backend & APIs: FastAPI · Python · REST APIs · WebSockets · Swagger · Postman · Docker

AI / GenAI: LLM Pipelines (GPT-4) · RAG · LangChain · FAISS · Vector Embeddings · Prompt Engineering · Multi-Turn Dialog · Agentic AI

Real-Time & Avatar: Unreal Engine 5 · MetaHuman · Unity Engine · Ready Player Me · SALSA LipSync · OBS Studio · VB-Cable · WebSocket Streaming

Cloud & DevOps: Azure OpenAI · Azure DevOps · Git/GitHub · CI/CD · Docker · AWS (Basic) · GCP (Basic)

Tools & Process: Jira · Agile/Scrum · Postman · Swagger

PROFESSIONAL EXPERIENCE

Ascendion · AI Engineer (Python / Voice Bot Developer)

December 2022 – Present | Bangalore, India

Voice Bot & Azure STT/TTS Development

- Architected and delivered 4 production-grade voice AI and chatbot systems for enterprise clients including Pfizer — covering real-time STT → LLM → TTS pipelines, live-meeting AI avatars, and RAG document Q&A.
- Engineered complete voice pipelines using Azure Speech-to-Text (STT), Azure Text-to-Speech (TTS), and Azure OpenAI GPT-4; implemented real-time transcription, chunked audio delivery, and streaming inference for low-latency voice interactions.
- Built Python FastAPI backend services exposing STT, LLM, and TTS endpoints; integrated voice bots with audio capture, real-time transcription, and AI-generated response playback over REST APIs and WebSockets.
- Reduced voice bot failure rate by ~25% through LLM timeout fallbacks, STT retry logic, and graceful text-only degradation — improving reliability in production deployments.

Microsoft Teams & Real-Time Voice Integration

- Engineered a real-time AI avatar system that joins Microsoft Teams and Zoom meetings as a live participant: captured remote speaker audio, routed through Azure STT → GPT-4 → ElevenLabs TTS → avatar playback with streamed responses.
- Implemented WebSocket-based streaming for low-latency GPT-4 token delivery and chunked audio playback, achieving natural real-time conversation cadence within live Teams/Zoom sessions.
- Configured audio routing stack using VB-Cable and OBS Studio as virtual audio/camera bridges to inject AI-generated voice and video output directly into Microsoft Teams as a standard meeting participant.
- Developed Unity C# and Python integration layers handling audio recording, API communication, TTS playback, and animation state management — translating directly to Teams Bot voice handling patterns.

Python Backend & API Development

- Designed and deployed RESTful FastAPI services for STT, LLM inference, TTS, and RAG endpoints; validated all APIs via Swagger and Postman across multiple enterprise client projects.
- Built production RAG pipelines with document chunking, Azure OpenAI embeddings, and FAISS vector indexing; improved retrieval accuracy by ~40–50% through chunk-size optimisation and semantic search tuning.
- Integrated LangChain for LLM orchestration, multi-turn conversation memory, and retrieval chains in Python backend services.

Delivery & Collaboration

- Collaborated in Agile/Scrum teams using Azure DevOps CI/CD, Git/GitHub, and Jira; delivered production systems across remote and on-site environments for global enterprise clients.

KEY PROJECTS

Real-Time AI Voice Bot for Microsoft Teams & Zoom — Python · FastAPI · Azure Speech STT/TTS · Azure OpenAI GPT-4 · ElevenLabs TTS · WebSockets · OBS · VB-Cable · Unreal Engine 5

- Built an AI voice participant that joins live Teams/Zoom meetings: captured remote audio → real-time Azure STT transcription → GPT-4 streamed response → TTS synthesis → voice playback in-meeting.
- Implemented WebSocket streaming pipeline for token-level GPT-4 delivery and chunked TTS audio output, minimising response latency in live meeting interactions.
- Configured OBS Studio and VB-Cable as virtual audio/video bridges to inject AI voice and avatar output into Teams as a fully functional meeting participant.

Pfizer Enterprise RAG Application — Python · FastAPI · Azure OpenAI GPT-4 · LangChain · FAISS · Vector Embeddings

- Built a Python-based RAG system enabling semantic Q&A over Pfizer's enterprise document library; improved response accuracy by ~40–50% over baseline through prompt engineering and chunking optimisation.
- Implemented full ingestion pipeline (parse → chunk → embed → FAISS index) and query pipeline (embed query → similarity search → GPT-4 context injection) using Python and FastAPI.

Voice Chatbot with Ready Player Me Avatar — Python · FastAPI · Azure STT/TTS · Azure OpenAI GPT-4 · Unity · C# · SALSA LipSync

- Built end-to-end voice chatbot: Python FastAPI backend with dedicated STT, GPT-4, and TTS endpoints + Unity C# client for audio capture, API calls, and real-time TTS playback with lip-sync animation.

HR Induction Portal — Voice Avatar Chatbot — Python · FastAPI · Azure STT/TTS · Azure OpenAI GPT-4 · FAISS · RAG · Unity · C#

- Developed on-demand HR onboarding voice bot; built RAG backend ingesting HR policies and FAQs into FAISS vector store for accurate context-aware Q&A via voice interface.

EDUCATION

Bachelor of Engineering — Computer Science · Visvesvaraya Technological University (VTU) 2020 | Hospete, India

AWARDS & RECOGNITION

- Circle Champion Award — recognised for outstanding contributions to AI and voice bot product delivery.
- Rookie Star Award — awarded for exceptional performance in early career tenure.